Supporting Information

Electrified cement production via anion-mediated electrochemical calcium extraction

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Methods

Preparation of catalysts

The anodes were prepared by depositing IrO_2 on a titanium felt substrate (Fuel Cell Store) followed by thermal decomposition. The titanium felt (8x8 cm^2) was rinsed thoroughly with acetone and DI water and etched with boiling 0.5 M oxalic acid for 30 min. The etched titanium felt was then spray coated on a hot plate held at 80 °C with a solution consisting of 200 mg $IrCl_3 \cdot xH_2O$ (99.8%, Alfa Aesar, USA) dissolved in 13 mL of ethanol. The titanium felt coated with $IrCl_3$ was then calcinated at 500 °C for 10 minutes and cut to size to obtain the final anode electrode.

For the hydrogen evolution reaction catalyst at the cathode, we deposited platinum supported carbon nano powder (Pt/C, 40 wt. % loading, Sigma-Aldrich) catalyst onto a hydrophilic carbon paper. The cathode electrodes were prepared via spray-deposition of an ink containing commercial Pt/C (40 wt.%) catalyst, a perfluorosulfonic acid ionomer (Aquivion® D79-25BS liquid, dispersion, 25 wt% in water, PFSA), and methanol until a Pt loading of 0.40 mg/cm^2 is achieved. The electrodes were dried under room conditions overnight prior to performance experiments.

Electrochemical measurements

The electrocatalytic experiments were conducted at ambient temperature in a membrane-electrode assembly (MEA) electrolyzer cell (316L stainless steel cathode, and grade 2 titanium anode) with an active area of 1 cm^2 accessed with a serpentine channel, illustrated in Fig. S3. Cation exchange membrane (CEM, Nafion 117, Fuel Cell Store), anion exchange membrane (AEM, Sustainion X37-50 Grade RT, Dioxide Materials), and a 0.4 mm thick flow channel were sandwiched between the IrO_2 anode and Pt-C cathode. 1 M $CaCl_2$ electrolyte was fed into the anode at a rate of 5 mL/min using a peristaltic pump. The produced HClO/HCl solution flowed back to the CO_2 removal unit at the same rate to dissolve limestone (99%, Sigma Aldrich). The cathode was circulated with 1 M NaOH electrolyte at a rate of 5 mL/min. For the middle chamber, a 0.4 mm (measured without compression) silicone flow channel was incorporated between CEM and AEM to collect the $Ca(OH)_2$ precipitation. 3 M NaCl electrolyte at a rate of 5 mL/min was circulated through the silicone flow channel. As gypsum (99%, Sigma Aldrich) is a zero-carbon feedstock and is soluble

in water, CO_2 removal unit is removed for these experiments. Electrochemical measurements were carried out using Autolab PGSTAT204 in a galvanostatic mode. The reported current densities are based on the geometric surface area of the electrode (1 cm^2). The current efficiency is calculated by using the difference in the amount of calcium in the system before and after 1 hr of continuous electrolysis. We measure the difference by using an inductively coupled plasma - optical emission spectrometry (method details in Note S2).

The CO_2 reduction experiments were performed in a manner as described in the previous work.[1,2] Cathodes were prepared by sputtering 200 nm of Cu (99.99%) onto a porous PTFE membrane followed spray depositing a CO_2 regeneration layer. Anode is IrO_2 on a titanium felt support prepared in a similar manner as described for the ClER. Collected CO_2 from the CO_2 removal unit was fed into the cathode, while the anode was circulated with 0.01 M H_2SO_4 electrolyte at a rate of 5 mL/min with a peristaltic pump.

Products analysis

SEM imaging and compositional analysis of the samples was conducted using a JEOL JSM-6500F equipped with an energy-dispersive X-ray detector (Oxford Instrument XMax 150 mm^2), operating at 10-kV accelerating voltage for imaging and 15 kV for energy-dispersive X-ray spectroscopy analysis. XRD patterns were collected using the Bruker AXS GmbH, D8 Advance, using Cu radiation.

The CO_2RR gas products, oxygen, and CO_2 were analyzed by sampling the gas outlet stream with a gas chromatograph (Perkin Elmer Clarus 590) coupled with a thermal conductivity detector (TCD) and flame ionization detector (FID). The gas chromatograph is equipped with a Molecular Sieve 5A Capillary Column and a packed Carboxen-1000 Column with argon as the carrier gas. The volumetric gas flow rates in and out of the cell were measured with a bubble column. The liquid products were quantified using proton nuclear magnetic resonance spectroscopy ($1H^+$ NMR) on an Agilent DD2 500 spectrometer in D_2O using water suppression mode, with dimethyl sulfoxide (DMSO) as the internal standard.

Belite and alite synthesis

Electrochemically precipitated $Ca(OH)_2$ was mixed with SiO_2 (99%, Sigma Aldrich) in a 3:1 molar ratio. The well-mixed powders were placed in platinum crucibles and heated at 2 ℃ per min to 1200 ℃ and 1500 ℃ in a tube furnace. The temperature was held at 1200 ℃ and 1500 ℃ for 2 hours, then the powders were furnace-cooled by turning off the powder. During the heating and cooling, no protective gas was used.

Note S1:

Technoeconomic analysis

We perform technoeconomic analysis of the decarbonized cement for both cases with limestone and gypsum as feedstock, respectively. The assumptions and model used to calculate the plant-gate levelized cost are provided below.

A list of assumptions is as follows:

1. The production capacity of the plant is 4100 tonne cement/day.
2. A current efficiency of 93%.
3. The electrolyzer capital cost is adopted from ref.[3] as \$4300/m^2 which is typically for a PEM electrolyzer.
4. The capital cost of conventional cement plant is \$18/tonne cement[4,5].
5. The electrolyzer lifetime is 20 years, which is commonly used for other electrochemical processes[3,6].
6. The renewable electricity price is assumed to be 2 cent/kWh.
7. The plant runs with a capacity factor of 0.9.
8. The discount rate is 10%.
9. For the catalyst and membrane replacement cost:
    - Platinum costs \$28/gram[7] and the cathode has a loading of 0.4 mg/cm^2.
    - Iridium cost \$50/gram[7] and the anode has a loading of 1 mg/cm^2.
    - The US National Renewable Energy Laboratory estimated the unit membrane cost for an electrolyzer with a single membrane to be \$500/m^2 (ref.[6]). The cement

electrolyzer uses two pieces of membrane and therefore, a unit membrane cost of $1000/m^2 is used for the analysis.

- The catalysts and membranes need to be replaced every 5 years.

10. The final cement product consists of 65% CaO by weight and 35% SiO_2 by weight.

- Clay which consists of 70% SiO_2 by weight costs $16/tonne[7].
- Limestone costs $12/tonne[7].
- Gypsum which consists of 75% $CaSO_4$ by weight costs $8.6/tonne[7].

11. Installation cost and balance of plant is 35% of total capital cost.

$$Total\ charge\ per\ tonne\ Ca(OH)_2\ (C/tonne\ Ca(OH)_2)$$

$$= \frac{1 \times 10^6\ (g/tonne)\ \times No.e\ transfer\ \times Faraday\ constant\ (C/mol)}{Ca(OH)_2\ molar\ mass\ (g/mol)\ \times\ coulombic\ efficiency}$$

$$Energy\ consumed\ per\ tonne\ Ca(OH)_2\ (in\ kWh/tonne\ Ca(OH)_2)$$

$$= \frac{charge\ per\ tonne\ Ca(OH)_2\ (C/tonne\ Ca(OH)_2)\ \times cell\ voltage\ (V)}{3.6 \times 10^6\ (J/kWh)}$$

$$Energy\ consumed\ per\ tonne\ cement\ (kWh/tonne\ cement)$$

$$= \frac{energy\ per\ tonne\ Ca(OH)_2\ (kWh/per\ tonne\ Ca(OH)_2) \times Ca(OH)_2\ molar\ mass\ (g/mol)}{CaO\ molar\ mass\ (g/mol)}$$

$$\times weight\ percent\ of\ CaO\ in\ cement\ (\%)$$

$$Electricity\ cost\ per\ tonne\ cement\ (\$/tonne\ cement)$$

$$= energy\ per\ tonne\ cement\ (kWh/tonne\ cement)$$

$$\times electricity\ price\ (\$/kWh)$$

$$Electrolyzer\ area\ required\ (m^2)$$

$$= \frac{charge\ per\ tonne\ Ca(OH)_2 \left(\frac{C}{tonne} Ca(OH)_2\right) \times \frac{Ca(OH)_2\ molar\ mass\ \left(\frac{g}{mol}\right)}{CaO\ molar\ mass\ \left(\frac{g}{mol}\right)} \times weight\ percent\ of\ CaO\ in\ cement\ (\%)}{capcity\ factor \times 24\ (hr/day) \times 3600\ (s/hr)}$$

$$\times\ plant\ capacity\ (tonne/day)\ \ /\ current\ density\ (A/m^2)$$

$$Electrolyzer\ cost\ (\$) = electrolyzer\ area\ per\ tonne\ cement\ (m^2) \times$$

$$unit\ area\ electrolyzer\ cost\ (\$/m^2)$$

$$Capital\ recovery\ factor = \frac{discount\ rate \times (1 + discount\ rate)^{lifetime}}{(1 + discount\ rate)^{lifetime} - 1}$$

*A 0.118 capital recovery factor is used for the electrolyzer.

$$Electrolyzer\ capital\ cost\ \ per\ tonne\ cement\ (\$/tonne\ cement)$$

$$= \frac{capital\ recovery\ factor\ \times electrolyzer\ capital\ cost\ (\$)}{capacity\ factor\ \times 365\ (day/yr)\ \times plant\ capacity\ (tonne/day)}$$

$$Catalyst\ replacement\ cost\ (\$)$$

$$= electrolyzer\ area\ required\ (m^2) \times catalyst\ loading\ (g/m^2)$$

$$\times catalyst\ price\ (\$/g)$$

$$Membrane\ replacement\ cost\ (\$)$$

$$= electrolyzer\ area\ required\ (m^2) \times membrane\ unit\ cost\ (\$/m^2)$$

*A 0.264 capital recovery factor is used for the catalyst replacement cost.

$$Catalyst\ and\ membrane\ replacement\ cost\ per\ tonne\ cement\ (\$/tonne\ cement)$$

$$= \frac{capital\ recovery\ factor\ \times catalyst\ \&\ membrane\ replacement\ cost(\$)}{capacity\ factor\ \times 365\ (day/yr)\ \times plant\ capacity\ (tonne/day)}$$

Note S2:

Energy analysis

Part 1. Energy analysis for the previous electrochemical cement synthesis method

We perform energy analysis for the electrochemical cement synthesis based on the performance achieved in previous work by Ellis et al.[8]

Below outlines the assumptions used for the analysis:

- A current efficiency of 85%.
- A full cell voltage of 3.5 V. The energy provided by the co-generated H_2 is sufficient to provide thermal energy requirement of the decarbonized cement approach (also see Supplementary Note 1 Part 2)
- The electrolyzer produces 0.5 tonne CO_2/tonne cement. The CO_2 is produced as a mixture with O_2 in a CO_2:O_2 molar ratio of 2:1. Amine-based capture is assumed to be use to capture and separate this mixture at an energy cost of 4 GJ/tonne CO_2.[9]
- The final cement product consists of 65% CaO by weight and 35% SiO_2 by weight.

The electricity input for the electrochemical process can be calculated as:

$$Total\ charge\ (in\ C)\ per\ tonne\ Ca(OH)_2$$

$$= \frac{1 \times 10^6\ (g/tonne)\ \times No.e\ transfer\ \times Faraday\ constant\ (C/mol)}{Ca(OH)_2\ molar\ mass\ (g/mol)\ \times\ coulombic\ efficiency}$$

$$Energy\ consumed\ (in\ kWh)\ per\ tonne\ Ca(OH)_2$$

$$= \frac{charge\ per\ tonne\ Ca(OH)_2(C)\ \times cell\ voltage\ (V)}{3.6 \times 10^6\ (J/kWh)}$$

$$Energy\ consumed\ (in\ kWh)\ per\ tonne\ cement = \frac{energy\ per\ tonne\ Ca(OH)_2 (kWh) \times Ca(OH)_2\ molar\ mass\ (g/mol)}{CaO\ molar\ mass\ (g/mol)} \times weight\ percent\ of\ CaO\ in\ cement\ (\%)$$

By inputting the values into the equations, the electricity requirement for the electrochemical cement synthesis method is calculated to be 9.2 GJ/tonne cement. The electrolyzer also outputs 0.5 tonne CO_2/tonne cement in a CO_2/O_2 mixture. Separating such a mixture requires amine-based capture at 4 GJ/tonne CO_2, adding an extra 2 GJ/tonne cement on top of the electricity requirement for the electrolyzer. Therefore, as shown in Fig. S1, the total energy input for the electrochemical cement synthesis method is estimated at 11.2 GJ/tonne cement – 2.4 times over the US averaged energy requirement for conventional cement production process (4.6 GJ/tonne cement[10]).

Part 2. Energy analysis for the proposed electrified method using anion-mediated electrochemical calcium extraction

For conventional cement process, portland cement production undergoes 2 steps: Calcination of $CaCO_3$ into CaO at ~ 900 °C (Eq.1) and sintering of CaO and SiO_2 at ~ 1500 °C (Eq.2).

Eq.1: $CaCO_3 \rightarrow CaO + CO_2$ $\Delta H°=178$kJ/mol

Eq.2: $3CaO + SiO_2 \rightarrow Ca_3SiO_5$

The thermal energy intensity of cement is averaged at 4.6 GJ/tonne cement which is ~35% of the total thermal energy is used for the sintering process at 1.6 GJ/tonne cement.[11]

In the proposed electrochemical cement unit, $Ca(OH)_2$ and H_2 are produced as feedstock and fuel for portland cement production, respectively. $Ca(OH)_2$ rather than limestone is then is calcinated

to CaO (Eq. 3) and the CaO is subsequently mixed with SiO_2 for sintering (Eq. 2).

Eq.3: $Ca(OH)_2 \rightarrow CaO + H_2O$ $\Delta H°=139kJ/mol$

Eq.2: $3CaO + SiO_2 \rightarrow Ca_3SiO_5$

The standard free energy of $Ca(OH)_2$ calcination is lower than that of limestone, which we calculated the thermal energy for Eq.3 to be 1.6 GJ/tonne cement. Using a similar energy input (1.6 GJ/tonne cement) as the sintering process for conventional cement process, the total thermal energy is estimated to be 3.2 GJ/tonne cement for the proposed electrified cement production method.

The energy provided by the H_2 fuel can be calculated as follows:

$$H_2\ production\ (mol)\ per\ tonne\ cement = \frac{charge\ transfer\ per\ ton\ cement(C)}{No.e\ transfer\ \times Faraday\ constant\ (C/mol)}$$

$$Energy\ provided\ by\ H_2\ per\ tonne\ cement$$

$$= H_2\ production\ per\ ton\ cement\ (mol) \times\ heating\ value\ (GJ/mol)$$

The energy provided by H_2 is calculated to be 4 GJ/tonne cement, in excess to provide thermal energy requirement of the decarbonized cement approach.

The electricity for the electrolyzer unit can be calculated using the similar approach outlined in Supplementary Note 2 Part 1. Using a current efficiency of 94% and a full cell voltage of 2.45V at 200 mA/cm^2, the electricity input for the electrified cement production via anion-mediated electrochemical calcium extraction is estimated at 5.9 GJ/tonne cement, comparable with conventional cement process (Fig. S1).

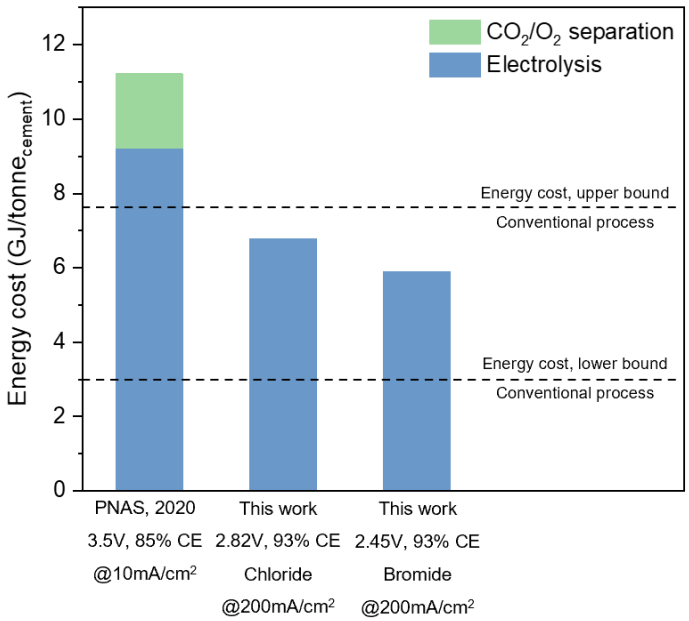

Fig. S1. Energy comparison. Cost to produce one tonne of cement by using the ECE process with previous literature.

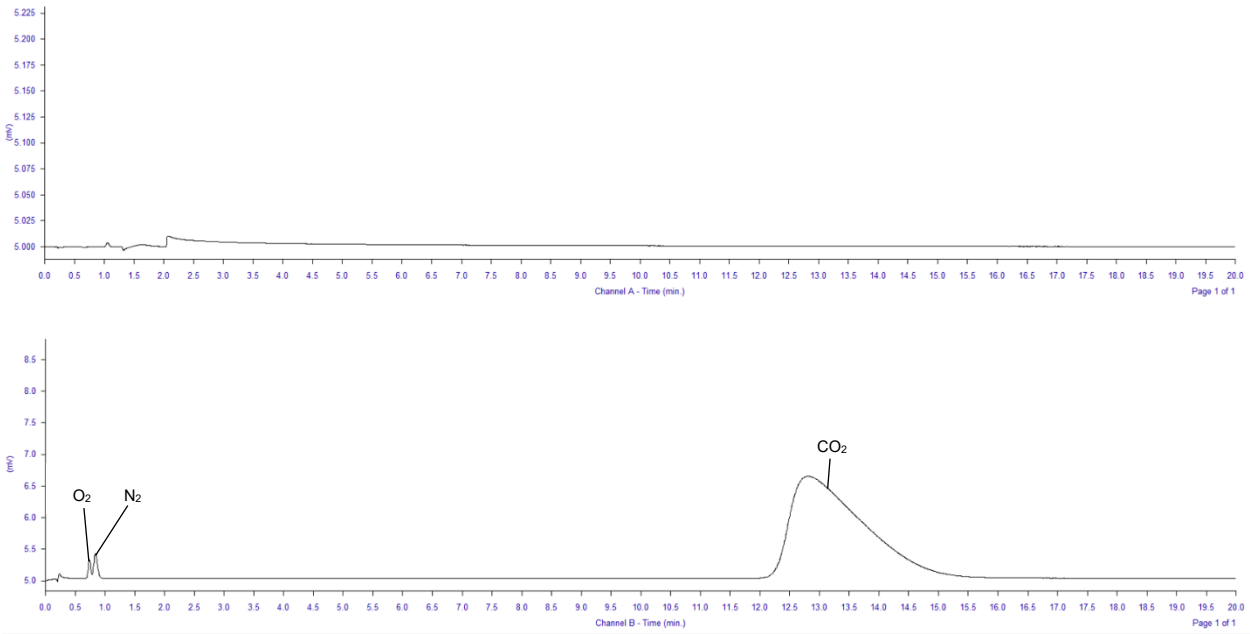

Fig. S2. Gas chromatography spectrum of the outlet gas from the CO_2 removal unit. The small amount O_2 and N_2 is from the ambient air.

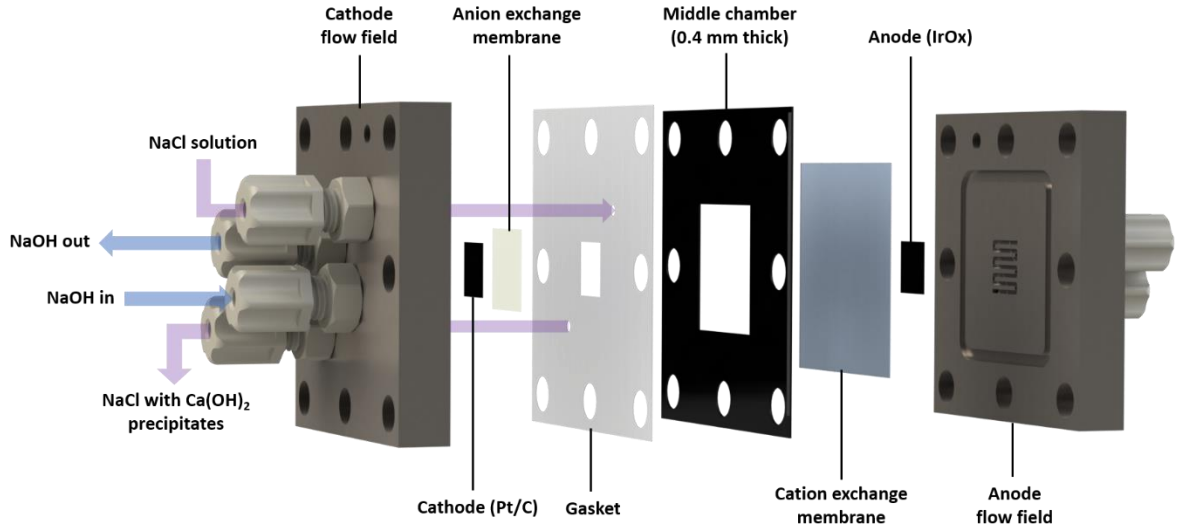

Fig. S3. Exploded view of the electrochemical unit.

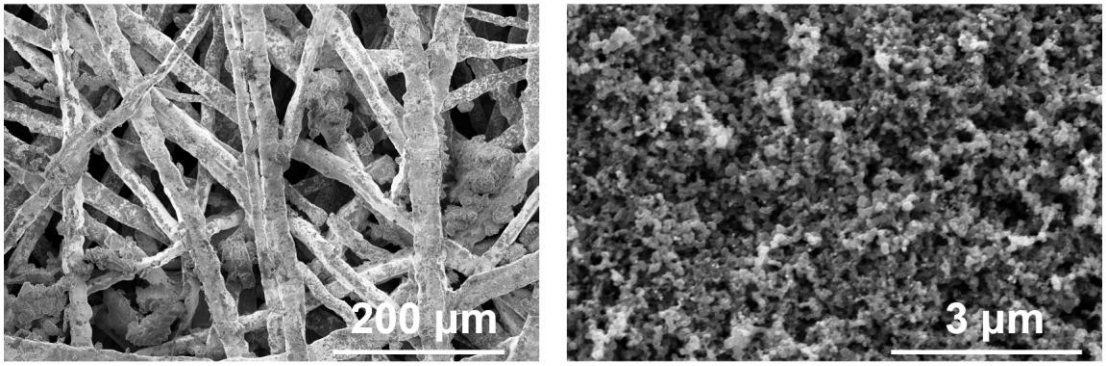

Fig. S4. SEM of the Pt/C. Surface images showing the nanostructured surface of Pt/C catalyst on carbon paper substrates.

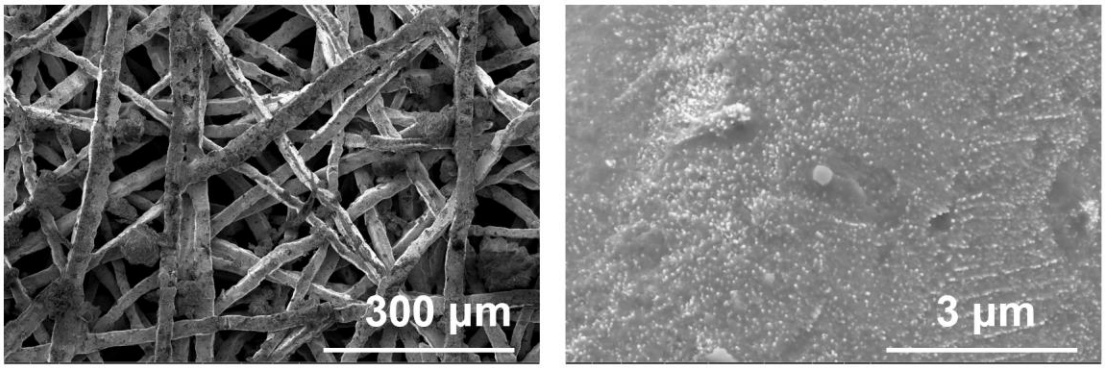

Fig. S5. SEM of the IrO_2. Surface images showing the nanostructured surface of IrO_2 catalyst on Ti felt substrates.

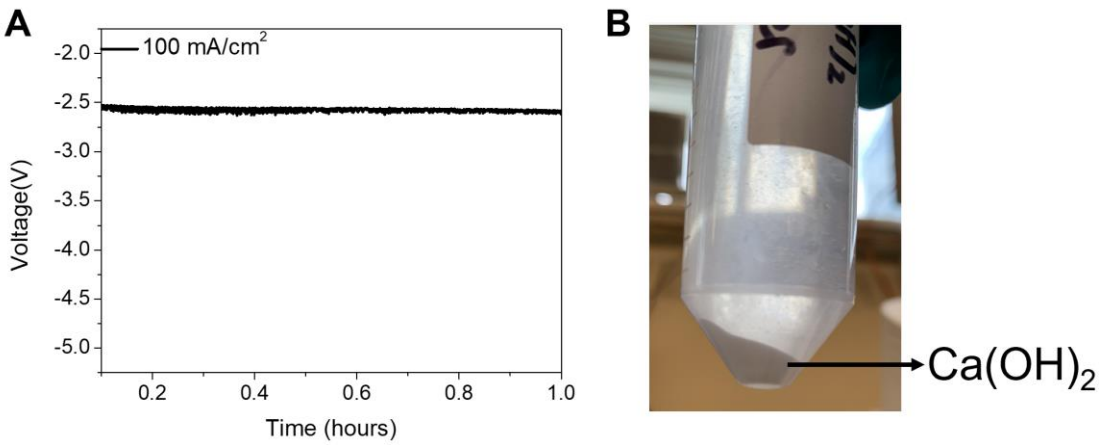

Fig. S6. Validation of $Ca(OH)_2$ electrosynthesis. (**A**) System voltage stability of electrosynthesis $Ca(OH)_2$ at 100 mA/cm^2, (**B**) The photograph of collected white precipitation from the middle chamber in a MEA electrolyzer.

Note S3:

Current efficiency calculation

The electrochemical calcium extraction process follows the following reactions:

$$HCl + HOCl + CaCO_3 \rightarrow \frac{1}{2}CaCl_2 + \frac{1}{2}Ca(OCl)_2 + CO_2 + H_2O$$

$$2Cl^- \rightarrow Cl_2 + 2e^-$$

$$Cl_2 + H_2O \leftrightarrow HOCl + HCl$$

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$$

$$Ca^{2+} + 2OH^- \rightarrow Ca(OH)_2$$

Therefore, the stoichiometric operation condition can be defined as for every two moles of electron transferred, one mole of $Ca(OH)_2$ is generated. This represents a current efficiency of 100%, and thus we express the current efficiency (CE) as:

$$CE = \frac{mole\ of\ Ca(OH)_2\ generated}{mole\ of\ electrons\ transferred/2}$$

However, we could not obtain an accurate measure of $Ca(OH)_2$ via weighing due to the following: The amount of $Ca(OH)_2$ precipitate produced at this scale is small, so direct gravimetric quantification introduced significant errors; and a small quantity of the $Ca(OH)_2$ precipitate remains on the ion exchange membranes. We weighted the amount of $Ca(OH)_2$ precipitates accumulated in the electrolyzer following 2 hour of electrolysis and found that the accumulated weight of $Ca(OH)_2$ collected was approximately ~80% of the theoretical weight of $Ca(OH)_2$ produced assuming 100% current efficiency for both the limestone and gypsum cases.

We therefore quantify the current efficiency instead by using two independent methods and take the average of the two methods as the reported current efficiency for the associated system.

Method 1:

We quantify the current efficiency by using the mole of calcium consumed in the electrolyte:

$$CE = \frac{mole\ of\ Ca^{2+}\ consumed\ in\ anolyte}{mole\ of\ electrons\ transferred/2}$$

The mole of calcium consumed can be calculated by using the difference in the amount of calcium in the anolyte before and after 2 hr of continuous electrolysis. We measure the difference by using an inductively coupled plasma - optical emission spectrometry (ICP-OES). To prepare the samples for the ICP-OES quantification, we first use an acid to dissolve all $CaCO_3$ into the solution. We then dilute the solution at least 100x prior to the analysis.

The current efficiency for the limestone and gypsum cases was 94%±3% and 85%±4%, respectively.

Method 2:

As protons generated by the electrolysis can either be presented in the anolyte (or to react with $CaCO_3$ for the limestone case) or transported through the CEM, we quantify the current efficiency as the percentage of protons generated that remain on the anode side.

$$CE = \frac{mole\ of\ H^{+}\ on\ the\ anode\ side}{mole\ of\ H^{+}\ generated}$$

For the limestone case, protons on the anode side can react with $CaCO_3$ to form CO_2, and we can use the measured CO_2 flow rate exiting the CO_2 removal unit to quantify the current efficiency. As shown in **Fig. S7**, the current efficiency is 91%±4% using this method.

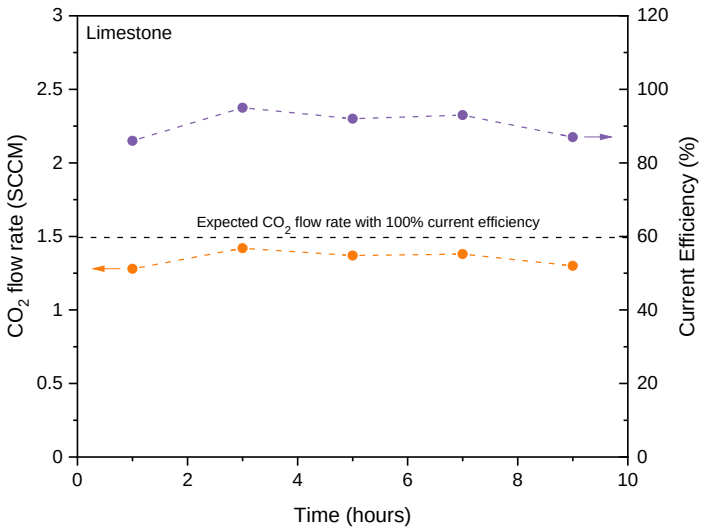

Fig. S7. CO_2 flow rate released from the CO_2 removal over 10 hours of electrolysis at 200 mA/cm^2 with limestone as the feedstock and the corresponding current efficiency values.

For the gypsum case, protons on the anode side (instead of reacting with $CaCO_3$) accumulates in the anolyte, and we can measure the anolyte pH over time to quantify the current efficiency. As shown in **Fig. S8.**, the average current efficiency over 2 hours of electrolysis is 81%.

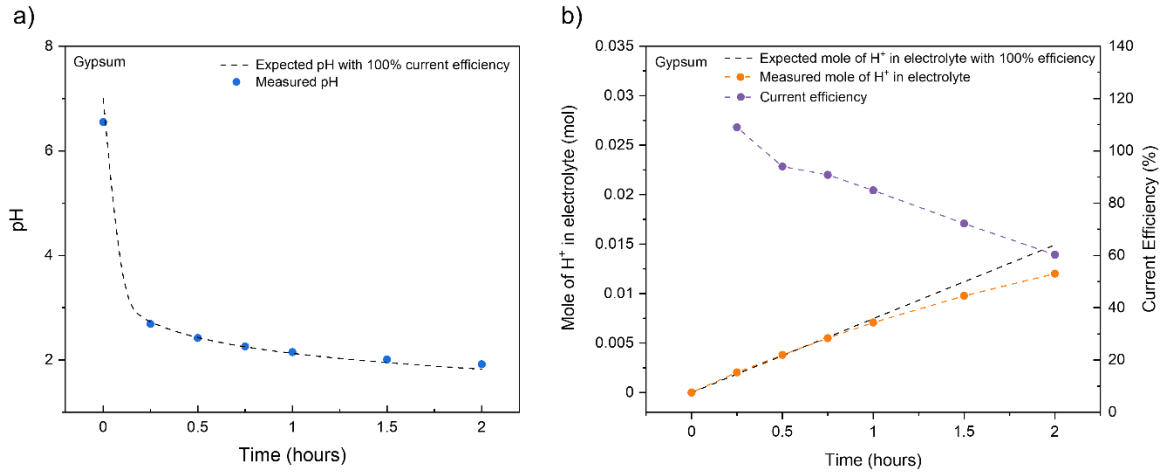

Fig. S8. A) the measured pH, and B) the measured accumulated mole of H^+ of anolyte over 2 hours of electrolysis at 200 mA/cm^2 with gypsum as the feedstock and the corresponding current efficiency values.

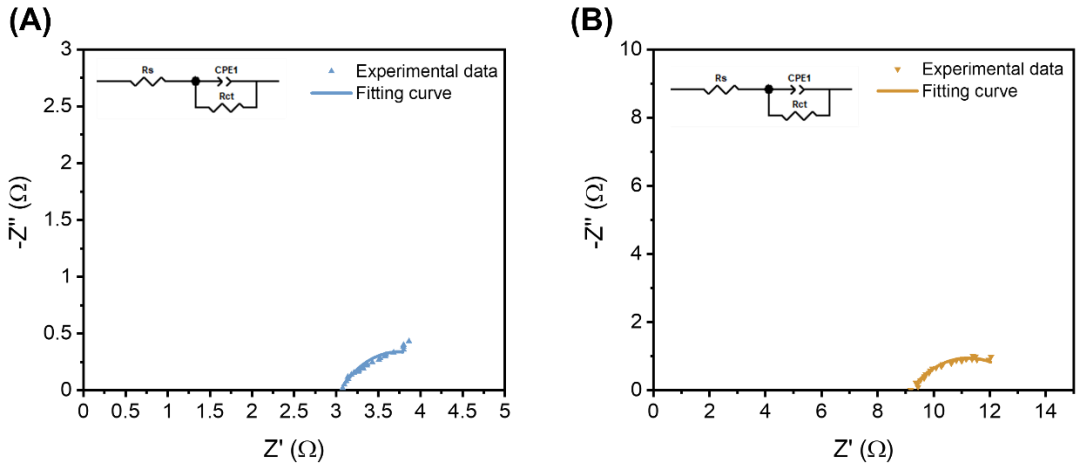

Fig. S9. Electrochemical impedance spectroscopy of the ECE process. (A) for using limestone as the feedstock, (B) for using gypsum as the feedstock.

The ohmic resistance and charge transfer resistance were measured through electrochemical impedance spectroscopy (EIS) in a two-electrode cell under a constant current density of 100 mA/cm^2, and the data points were obtained between 10 Hz and 100 kHz. The simplified equivalent circuit (insert Figure S9) was chosen to model the cell. The series resistant (R_s) expresses the resistance of electrodes, the resistance of conductive plates, and the resistance of membrane and ionic transport. The charge transfer resistance (R_{ct}) models the double layer of charge at the interface between the membrane and electrode. The detailed fitting parameters are listed in Table S1. In both cases, the series resistances are dominant. This finding is consistent with the voltage trend shown in Fig. 2b and 3b; the applied voltages show linear trends with respect to current density. The gypsum case shows a higher series resistance and charge transfer resistance, and we associate the higher voltage with the lower solubility of gypsum (2.0-2.5 g/L at 25°C).

	R_s (Ω cm^2)	Error (Ω cm^2)	R_{ct} (Ω cm^2)	Error (Ω cm^2)
Limestone	3.05	0.02	1.37	0.17
Gypsum	9.16	0.07	4.22	0.28

	Ca^{2+}	Na^{+}	Cl^{-}
Concentration	190 ppm	0.70 ppm	0.91 ppm

Table. S1. Elemental analysis of the precipitates by inductively coupled plasma optical emission spectroscopy. The precipitates were rinsed by DI water and dried in vacuum prior to the analysis. By assuming all calcium is associated with $Ca(OH)_2$ and all chloride is associated with NaCl and the final cement weight consisting of 65% CaO by weight, the chloride concentration is calculated to be ~0.2 wt% of the cement weight below the threshold standard set for chloride content in the final cement product.

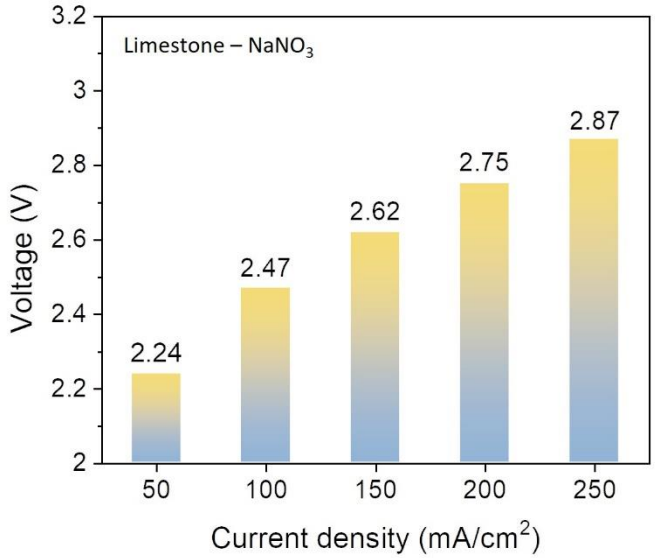

Fig. S10. Voltage using $NaNO_3$ as the flowing electrolyte. Voltage of the electrochemical unit at different current densities using $NaNO_3$ as the flowing electrolyte in the middle chamber and limestone as the feedstock.

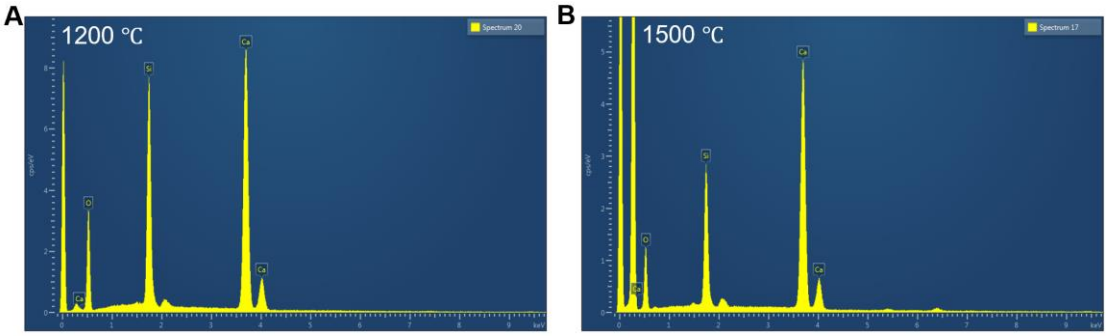

Fig. S11. Energy-dispersive X-ray of the cement. Energy-dispersive X-ray element ratio of prepared phases at sinter temperature of (**A**) 1200 ℃ and (**B**) 1500 ℃.

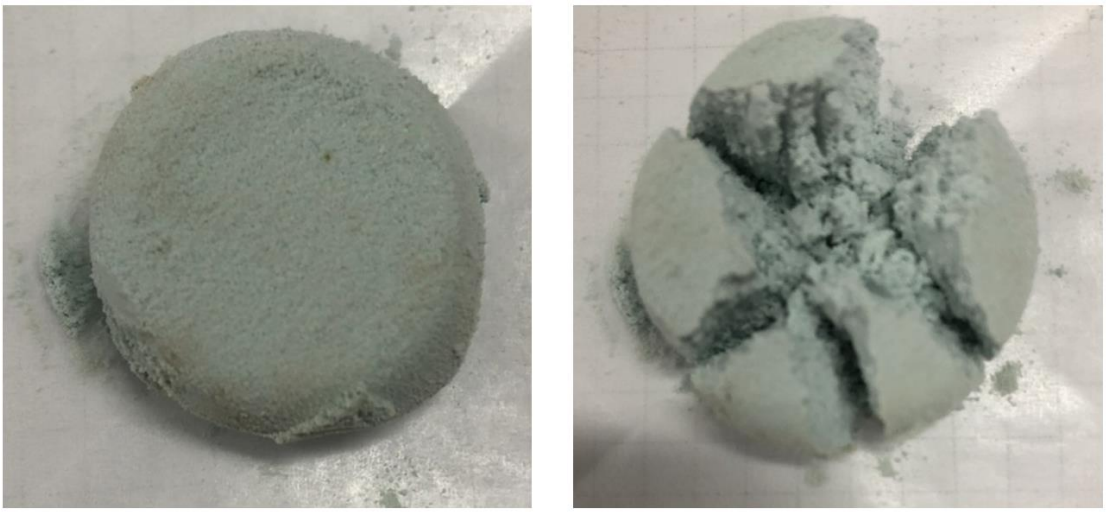

Fig. S12. Photographs of prepared alite, shows grey green color.

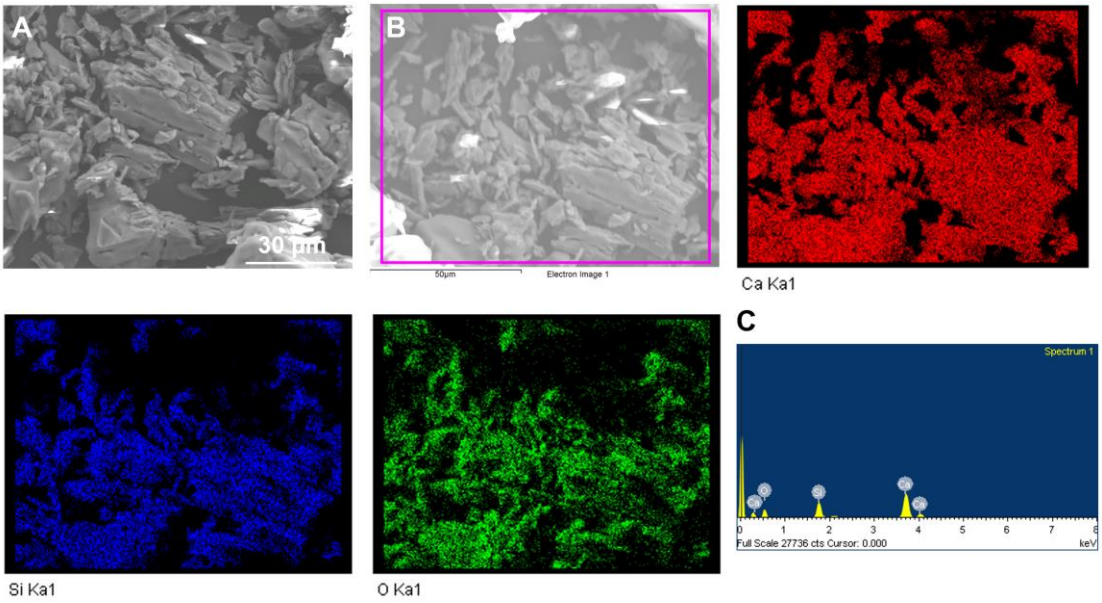

Fig. S13. Characterizations of the prepared alite. (**A**) Surface morphology, (**B**) composition maps and (**C**) Energy-dispersive X-ray element ratio of prepared alite. Scale bar is 30 μm.

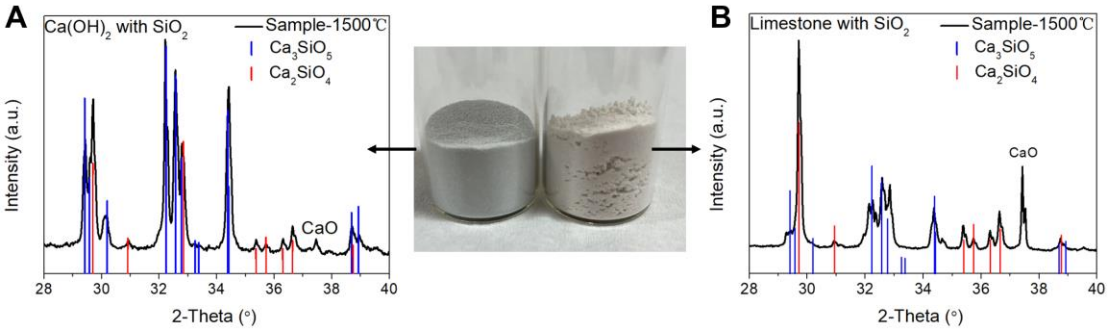

Fig. S14. XRD comparison of the cement produced by sintering (**A**) $Ca(OH)_2$ or (**B**) limestone with SiO_2 at the same calcinate condition. Middle figure is the resulting powders color comparison. The $Ca(OH)_2$ prepared sample is light grey color and limestone prepared sample is off-white color.

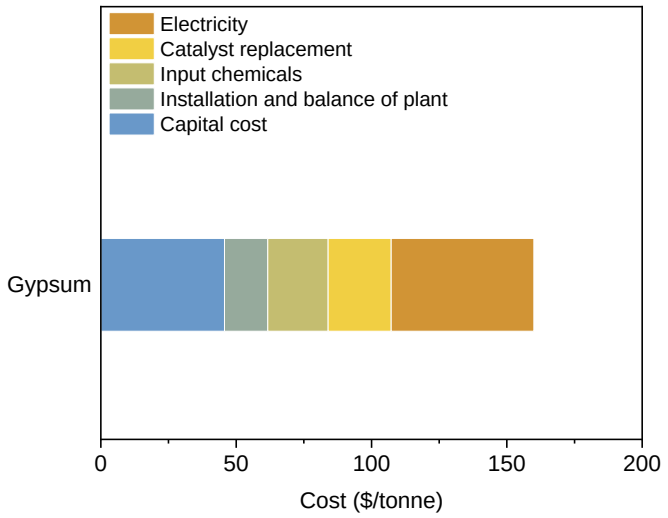

Fig. S15. TEA of the ECE process by using gypsum as feedstock. This TEA excludes cost to treat the acidic electrolyte.

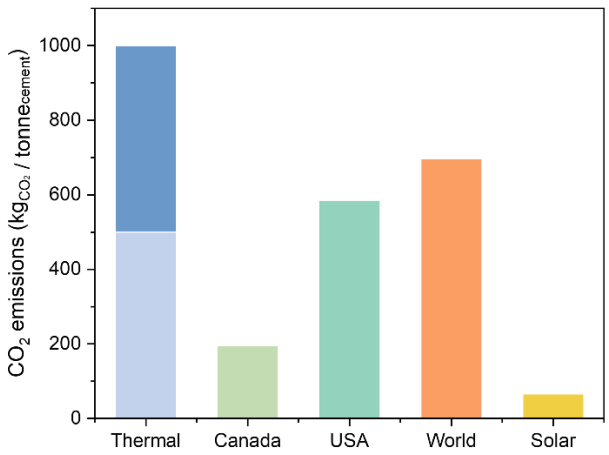

Fig. S16. Life cycle analysis of CO_2 emissions for the conventional thermal production process and the ECE process with different grid emission intensities (Canada = 0.119 kg CO_2/kWh, USA = 0.357 kg CO_2/kWh, World = 0.425 kg CO_2/kWh, and solar energy = 0.04 kg CO_2/kWh).

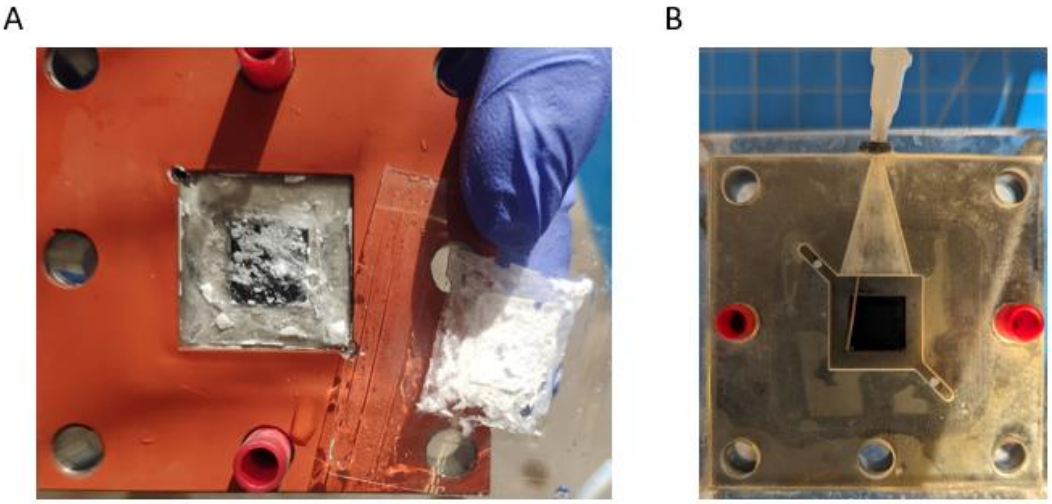

Fig. S17. Device for the stability test. (**A**) $Ca(OH)_2$ solid precipitates accumulated in the electrolyzer over the stability test. (**B**) Custom-designed middle chamber flow field and wiper insert for *in-operando* cleaning.

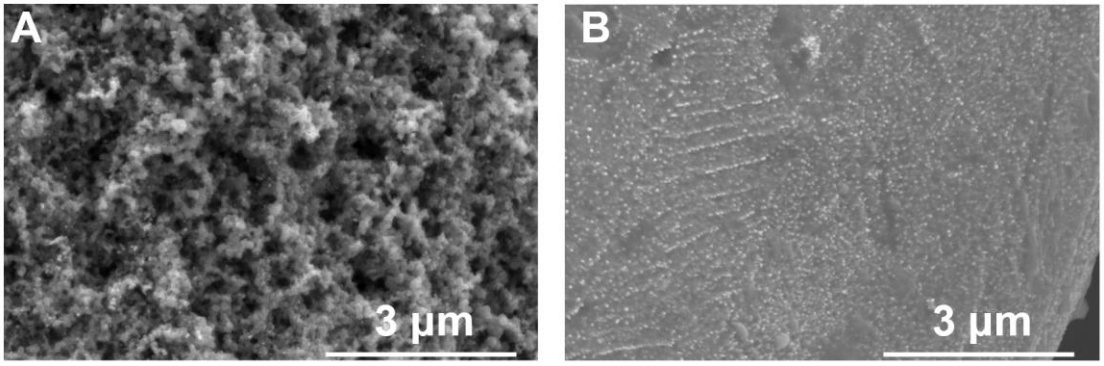

Fig. S18. Post-stability SEM images. (**A**) Pt/C cathode catalysts. (**B**) IrO_2 anode catalysts.

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